Apple Platform Deployment

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Intro to certificate management for Apple devices

Apple devices support digital certificates and identities, giving your organization streamlined access to corporate services. These certificates can be used in a variety of ways. For example, the Safari browser can check the validity of an X.509 digital certificate and establish a secure session with up to 256-bit AES encryption. This involves verifying that the site's identity is legitimate and that communication with the website is protected to help prevent interception of personal or confidential data. Certificates can also be used to guarantee the identity of the author or "signer" and to encrypt mail, configuration profiles, and network communications.

Actively managing which certificates your devices trust reduces the risk of a compromised or fraudulent Certificate Authority being used to intercept sensitive communications. This is especially important on unsupervised devices, where users can otherwise choose to trust certificates you haven't vetted.

Use certificates with Apple devices

Apple devices include a number of preinstalled root certificates from various Certification Authorities (CAs), and iOS, iPadOS, macOS, and visionOS validate the trust for these root certificates. These digital certificates can be used to securely identify a client or server, and to encrypt the communication between them using the public and private key pair. A certificate contains a public key, information about the client (or server), and is signed (verified) by a CA.

If iOS, iPadOS, macOS, or visionOS can't validate the trust chain of the signing CA, the service encounters an error. A self-signed certificate can't be verified without user interaction. For more information, see the Apple Support article List of available root certificates in iOS 26, iPadOS 26, macOS 26, tvOS 26, visionOS 26 and watchOS 26.

iPhone, iPad, Mac, and Apple Vision Pro devices can update certificates wirelessly (and for Mac, over Ethernet) if any of the preinstalled root certificates become compromised. You can turn off this feature using the device management restriction "Allow automatic updates to certificate trust settings," which prevents certificates updates over wireless or wired networks.

Supported identity types

A certificate and its associated private key are known as an *identity*. Certificates can be freely distributed, but identities need to be kept secure. The freely distributed certificate, and especially its public key, are used for encryption that can be decrypted only by the matching private key. The private key part of an identity is stored as a PKCS #12 identity certificate (.p12) file and encrypted with another key that's protected by a passphrase. An identity can be used for authentication (such as 802.1X EAP-TLS), signing, or encryption (such as S/MIME).

The certificate and identity formats Apple devices support are:

- *Certificate:* .cer, .crt, .der, X.509 certificates with RSA keys
- *Identity:* .pfx, .p12

Certificate trust

If a CA issues a certificate and its root isn't in the list of trusted root certificates, iOS, iPadOS, macOS, or visionOS don't trust the certificate. This is often the case with enterprise-issuing CAs. To establish trust, use the method described in certificate deployment. This sets the trust anchor at the certificate that's deploying. For multitiered public key infrastructures, it may be necessary to establish trust not only with the root certificate, but also with any intermediates in the chain. It's recommended you configure enterprise trust in a single configuration profile that you can update with your device management service as needed without affecting other services on the device.

Note: To perform automatic trust of additional root certificates you need to use a device management service.

Root certificates on iPhone, iPad, and Apple Vision Pro

Root certificates installed manually on an unsupervised iPhone, iPad, or Apple Vision Pro through a profile display the following warning:

"Installing the certificate *"name of certificate"* adds it to the list of trusted certificates on your *iPhone* or *iPad*. This certificate won't be trusted for websites until you enable it in Certificate Trust Settings."

The user can then trust the certificate on the device by going to Settings > General > About > Certificate Trust Settings.

Note: Root certificates on supervised devices, or that a device management service installs, turn off the option to change the trust settings.

Root certificates on Mac

Certificates installed manually through a configuration profile needs to have an additional action performed to complete the installation. After the profile is added, the user can navigate to System Settings and confirm the install.

The user can then review the details, cancel, or proceed by clicking Install. The user may need to provide a local administrator user name and password.

Note: For a Mac with macOS 13 or later, by default, macOS doesn't mark root certificates that you manually install with a configuration profile as trusted for TLS. If necessary, you can use the Keychain Access app to enable TLS trust. Root certificates on supervised devices, or that a device management service installs, turn off the option to change the trust settings, and macOS trusts them for use with TLS.

Intermediate certificates on Mac

Intermediate certificates are issued and signed by the Certificate Authorities' root certificate and they can be managed on a Mac using the Keychain Access app. These intermediate certificates have a shorter expiration date than most root certificates and are used by organizations so web browsers trust websites associated with an intermediate certificate. Users can locate expired intermediate certificates by viewing the System keychain in Keychain Access.

S/MIME certificates on Mac

If a user deletes any S/MIME certificates from their keychain, they can no longer read previous email that

was encrypted using those certificates.

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See also

Certificates device management payload settings for Apple devices

Certificate Preference device management payload settings for Apple devices

Certificate Revocation device management payload settings for Apple devices

Certificate Transparency device management payload settings for Apple devices

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Deploy Managed Device Attestation

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Distribute certificates

Helpful? Yes No

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